# Teach, Check, Respond

Session 1 — Find the Percent of a Number · Session 2 — Compare Percents · Maryland 6.AT.4

## LESSON ROUTE

*about 45 minutes per session*

| | |
|---|---|
| **Warm up · 5 min** | Warm-Up page (Session 1). Note who needs the prerequisite sheet. |
| **Launch · 7 min** | Slides: name the target. Students predict an answer to the Model, Reason, Explain task. |
| **Model · 10 min** | Guided Notes page for the session. Draw the deck's model next to the worked example. |
| **Practice · 18 min** | Practice Form A: core #1, #2, #4, #6 and Find the Error. Add more from what you see. |
| **Check · 5 min** | Exit Ticket. Sort into got it / almost / need help with the table below. |

*Shorter period: keep the model and the exit ticket; trim practice. Form B is for reteach or retake, so keep it unseen until then.*

## SESSION 1 · FIND THE PERCENT OF A NUMBER

*I can find the percent of a number exactly.*

### EXIT TICKET – WHAT TO LOOK FOR AND WHAT TO SAY NEXT

| # | Expected answer | Next prompt if it is not there yet |
|---|---|---|
| 1 | a. 36  b. 30 | You found 40% of 90 and 5% of 600. Find 10% of each whole first (9 and 60). How do 4 × 9 and half of 60 give the same answers? |
| 2 | 0.06 × 45 = $2.70 | Is the 6% tax on $45 closer to $2.70 or $27? Write 6% as a decimal (0.06, not 0.6) and explain. |
| 3 | **C** (0.15) | Why is 15 × 60 = 900 far too big for 15% of 60? Write 15% as 0.15 and find the part. |

### THE ERROR TO WATCH FOR

Multiplies by the percent number instead of its decimal (35 × 360 = 12,600), or writes a one-digit percent as tenths (6% as 0.6), so the part is bigger than the whole or 10 times too big.

**Reteach** Have the student find 10% of the whole on a 10-part percent bar, build the percent from it, and check that the part is less than the whole before multiplying by the decimal.

- Multiplying by the percent number (30 × 70) instead of the decimal (0.30 × 70).
- Writing small percents with the wrong place value (6% as 0.6, 9% as 0.9).

Calculation slip only: use Session 1 Prerequisite Practice A, then go back to the model. Wrong whole or wrong model: label the whole (100%) before redoing the arithmetic.

### FRESH CHECK – GIVE AFTER RETEACHING (STUDENTS HAVE NOT SEEN IT)

| | |
|---|---|
| **Check** | A library got 220 new books. 15% are mysteries. How many are mysteries? Use 10% and 5%, then check with an equation. |
| **Answer** | 10% of 220 = 22 and 5% = 11, so 15% = 22 + 11 = 33 mysteries. Check: 0.15 × 220 = 33. |

### SCORE THE CHECK

| | |
|---|---|
| **3** | Model fits the numbers, correct answer with units, explanation matches. Move on; remove a support next time. |
| **2** | Right setup with a small slip, or a thin explanation. Quick feedback, then one more new problem. |
| **1** | Some correct facts, but the wrong whole or model. Reteach with the session's model and the prerequisite page. |
| **0** | Not enough to tell yet. Ask the student to point, draw or say it before writing. |

### LANGUAGE ACCESS

Point and draw, then say it with a partner ("Partner A: "The whole is ___. 10% of it is ___, so 35% is ___." Partner B: "Check with an equation: ___ × ___ = ___. Is the part less than the whole?""), then write one sentence (Explain Your Thinking sheet), then explain alone. Fade one support each time.

## SESSION 2 · COMPARE PERCENTS

*I can compare two records fairly by converting each one to a percent.*

### EXIT TICKET – WHAT TO LOOK FOR AND WHAT TO SAY NEXT

| # | Expected answer | Next prompt if it is not there yet |
|---|---|---|
| 1 | a. 90%  b. 35% | 27 out of 30 and 7 out of 20: what do you multiply each bottom number by to get 100? Write each record as ___ out of 100. |
| 2 | a. 15/20 = 75%  b. 21/30 = 70%  c. Team A | Team B won more games (21) than Team A (15). Use 75% and 70% to explain why Team A still has the better record. |
| 3 | **B** (22 out of 25) | 43 out of 50 has the biggest top number. Write 17/20, 22/25, 43/50 and 8/10 out of 100. Which is greatest, and why is it not 43 out of 50? |

### THE ERROR TO WATCH FOR

Ranks records by the count made (36 > 32) and ignores the number of tries (60 vs. 40), or compares the part to 100 instead of to its own whole (33 of 55 called 33%).

**Reteach** Draw two equal 10-part bars, shade each player's makes out of their own tries, and ask which bar is fuller.

- Ranking records by the part alone (22 out of 40 "beats" 12 out of 15).
- Comparing the part to 100 instead of to its own whole (33 out of 55 called 33%).

Calculation slip only: use Session 2 Prerequisite Practice A, then go back to the model. Wrong whole or wrong model: label the whole (100%) before redoing the arithmetic.

### FRESH CHECK – GIVE AFTER RETEACHING (STUDENTS HAVE NOT SEEN IT)

| | |
|---|---|
| **Check** | Zoe got 19 of 20 quiz questions right. Max got 46 of 50 right. Who did better? Show both percents. |
| **Answer** | Zoe: 19/20 = 95/100 = 95%. Max: 46/50 = 92/100 = 92%. Zoe did better, even though Max got more questions right. |

### SCORE THE CHECK

| | |
|---|---|
| **3** | Model fits the numbers, correct answer with units, explanation matches. Move on; remove a support next time. |
| **2** | Right setup with a small slip, or a thin explanation. Quick feedback, then one more new problem. |
| **1** | Some correct facts, but the wrong whole or model. Reteach with the session's model and the prerequisite page. |
| **0** | Not enough to tell yet. Ask the student to point, draw or say it before writing. |

### LANGUAGE ACCESS

Point and draw, then say it with a partner ("Partner A: "___ out of ___ is ___%, and ___ out of ___ is ___%." Partner B: "So who did better? Why doesn't the bigger count decide it?""), then write one sentence (Explain Your Thinking sheet), then explain alone. Fade one support each time.